

Interview Q&A: pandas

1. How does pandas handle missing values when performing groupby operations?

pandas will propagate NaNs to the result of the groupby operation, unless a specific aggregation function is specified.

2. What is the difference between `pandas.DataFrame.groupby()` and `pandas.Series.groupby()`?

The main difference is that `DataFrame.groupby()` returns a `GroupBy` object, which can be used to perform aggregation operations on multiple columns, whereas `Series.groupby()` returns a `SeriesGroupBy` object, which can only be used to perform aggregation operations on the grouped series.

3. How to reset the index of a DataFrame while keeping the original index as a column?

You can use the `reset_index()` method with the `drop=False` parameter.

4. How to use pandas' `merge` function with multiple join conditions?

You can pass a dictionary of merge conditions to the `on` parameter, where each key is a column name and its corresponding value is another column name or a list of column names.

5. What is the difference between `pandas.DataFrame.shift()` and `pandas.DataFrame.resample()`?

The `shift()` function shifts the values of a DataFrame by a specified number of periods, while `resample()` groups data into bins based on a time interval.

6. How to use the `pandas.DataFrame.apply()` method with multiple functions?

You can pass a dictionary of functions to apply to different columns using the `apply()` method.

7. How to use the `pandas.DataFrame.melt()` function to unpivot a DataFrame?

The `melt()` function is used to unpivot a DataFrame from wide format to long format, allowing for easier analysis of data with multiple variables.